

Characteristic-Space Metrics in Factor Models

Identification and Inference

Mingyang Liu

Imperial Business School
Imperial College London

September 2026

From factor estimation to interpretable exposure comparisons

The starting point: inference for latent factor models

Bai (2003): How precisely can we estimate factors, loadings, and their common component in a large panel?

$$x_{it} = \lambda_i' f_t + e_{it}.$$

Here x_{it} is an observed panel outcome, f_t a vector of common factors, λ_i its unit-specific loading, and e_{it} a disturbance.

- Principal components estimate the latent common structure.
- Large-panel asymptotics deliver rates and limiting distributions.
- Rotations must be handled when interpreting factors and loadings.

The foundation: attach uncertainty to an estimated factor structure.

Bai (2003), *Econometrica* 71, 135–171. The large- N , large- T framework is the comparison here; the paper also discusses a large- N , fixed- T special case.

IPCA links changing exposures to observed characteristics

Instrumented Principal Components Analysis (IPCA) uses firm characteristics to explain time-varying factor loadings.

$$r_{i,t+1} = \beta'_{i,t} f_{t+1} + \varepsilon_{i,t+1}, \quad \beta_{i,t} = \Gamma'_\beta z_{i,t}.$$

Here $r_{i,t+1}$ is an excess return; $z_{i,t}$ contains characteristics known at t ; Γ_β is a common loading map.

- **Kelly, Pruitt, and Su (2019):** use this structure to study the cross section of risk and return, allowing characteristic intercepts.
- **Kelly, Korsaye, Pruitt, and Su (2026):** develop IPCA estimation and asymptotic theory for dynamic loadings in large panels.

Characteristics explain how a stock's exposure changes.

KPS (2019), *Journal of Financial Economics* 134, 501–524; KKPS (2026), *Instrumented Principal Component Analysis*, working paper, April 21 version.

A complementary research question

Framework	Central object	Question addressed
Bai (2003)	Latent factors and loadings	How can large panels support factor-model inference?
IPCA	Characteristic-driven loadings	How do observable instruments organize changing exposures?
This paper	Metric-indexed loading class	How is exposure allocated, and how precisely is that allocation measured?

The new issue arises when latent factors coexist with blocks that retain economic labels, and the rule separating those blocks is estimated.

An extension of the inferential object, not a replacement for IPCA.

Research roadmap. The IPCA row combines the empirical motivation in KPS (2019) and the theory in KKPS (2026).

Three results, three empirical uses

	Result	What it enables
Attribution	A declared metric selects a decomposition of a common fitted index.	Compare loading blocks under an explicit measurement rule.
Interior inference	The complete influence includes return and metric uncertainty.	Estimate standard errors for basis-independent exposure reports.
Boundary inference	Learning the metric also moves an exposure constraint.	Construct intervals valid from slackness to binding.

Separate what the model fits from what the researcher reports.

Main manuscript, Sections 3, 5–7; proofs and feasible constructions in Appendices A–D.

The model: latent, observed-factor, and intercept blocks

$$r_{i,t+1} = \underbrace{z'_{i,t}\Gamma_{\beta}f_{t+1}}_{\text{latent factors}} + \underbrace{z'_{i,t}\Gamma_{\delta}g_{t+1}}_{\text{observed factors}} + \underbrace{z'_{i,t}\Gamma_{\alpha}}_{\text{retained intercept}} + \varepsilon_{i,t+1}.$$

- N_t stocks at date t ; L characteristics in $z_{i,t} \in \mathbb{R}^L$.
- K latent factors f_{t+1} and K_o observed factors g_{t+1} .
- $\Gamma_{\beta} \in \mathbb{R}^{L \times K}$, $\Gamma_{\delta} \in \mathbb{R}^{L \times K_o}$, and $\Gamma_{\alpha} \in \mathbb{R}^L$ map characteristics into the three loading blocks.

The **complete fitted index** is $h_{t+1} = \Gamma_{\beta}f_{t+1} + \Gamma_{\delta}g_{t+1} + \Gamma_{\alpha}$; the fitted stock return is $z'_{i,t}h_{t+1}$.

We study the loading class with L, K, K_o fixed and $T \rightarrow \infty$.

Main manuscript, Section 2. The cross-sectional size N_t need not diverge; structural interpretation requires the conditions discussed below.

The metric measures overlap in stock exposures

Let Z_t stack stocks' characteristic vectors. Define

$$W_t = \frac{Z_t' Z_t}{N_t}, \quad Q_Z = \mathbb{E}(W_t), \quad \hat{Q} = \frac{1}{T} \sum_{t=1}^T W_t.$$

For two characteristic-loading directions $a, b \in \mathbb{R}^L$,

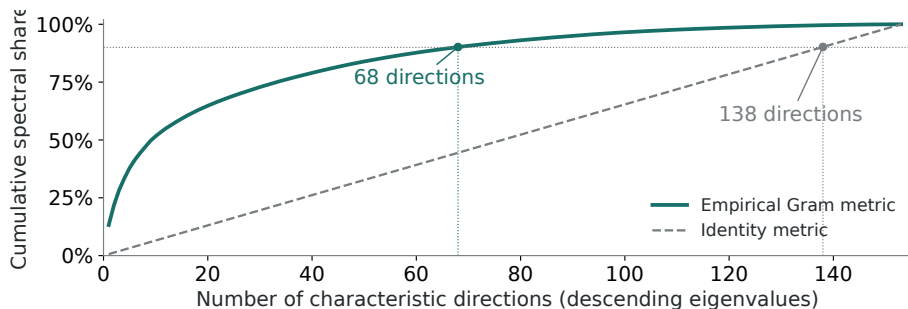
$$a' Q_Z b = \mathbb{E} \left[\frac{(Z_t a)' (Z_t b)}{N_t} \right].$$

- $a'b$ compares recorded coefficients in Euclidean units.
- $a' Q_Z b$ compares the stock exposure patterns those coefficients generate.
- Two value measures can be different coordinates yet emphasize similar stocks.

The Gram metric makes this overlap part of the attribution rule.

Main manuscript, Section 2. Q_Z is a second-moment matrix, not necessarily a centered covariance matrix; assume $Q_Z \succ 0$.

Recorded characteristics do not have identity geometry



153 U.S. equity characteristics, 1962–2025. The first five directions account for **37.9%** of spectral mass.

Rank scaling does not remove overlap between characteristic exposures.

Main manuscript, Table 1 and Figure 1; characteristic definitions follow Jensen, Kelly, and Pedersen (2023). Monthly ranks rescaled to $[-1/2, 1/2]$, missing values set to zero; trace-normalized average Gram matrix. This is characteristic geometry, not a count of priced risks.

A matched change of metric preserves fitted returns

For a declared positive-definite Q , select a representative satisfying

$$\Gamma'_\beta Q \Gamma_\beta = I_K, \quad \Gamma'_\beta Q [\Gamma_\delta, \Gamma_\alpha] = 0.$$

- Normalize latent loading units in the chosen metric.
- Remove latent-span overlap from retained loading blocks.
- Offset that reallocation in the latent factor realizations.

Q_1 representative or Q_2 representative



same complete h_{t+1}
same fitted returns and residuals

Residual fit alone cannot choose between these matched allocations.

Metric-section proposition, Main Section 3; explicit transformation in Appendix A and backup slide 1. A cap on intercept exposure can restrict the feasible fitted-index set.

The metric defines the object; curvature identifies it locally

Write $\theta = (\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha)$. The class $[\theta]$ treats latent rotations as equivalent. Let $\eta \in \mathbb{R}^d$ give d coordinates for locally distinct classes.

$$\tilde{Q}_T(\eta; Q) = \frac{1}{T} \sum_t a_t \min_f \frac{\|r_{t+1} - Z_t\{\Gamma_\beta f + \Gamma_\delta g_{t+1} + \Gamma_\alpha\}\|^2}{2N_t}.$$

The loading blocks are evaluated at (η, Q) ; a_t are fixed date weights with $T^{-1} \sum_t a_t = 1$. The displayed loss is equivalent up to a data-only constant.

- $\eta = 0$ is stationary for the population profiled criterion.
- Any cap on intercept exposure is slack and the quotient Hessian $H \succ 0$.
- Then nearby distinct classes have a strictly larger criterion value.

Normalization and statistical identification perform different tasks.

Main Sections 3–4. Exact profiling minimizes over date-specific factors; smooth charts, profile rank, and local branch selection are required.

When does the criterion target have a structural meaning?

- $[\theta_0]$: the class selected by the population profiled criterion.
- $[\theta^\circ]$: the class generating returns in the structural model.

With a fixed cross section, estimated factors absorb some of the same noise that enters the loading score. Mean-zero noise alone need not remove this effect.

Let $u_{t+1} = Z'_t \varepsilon_{t+1} / N_t$ be the managed disturbance. A sufficient covariance-alignment condition is

$$\mathbb{E}(u_{t+1} \mid \mathcal{C}_t) = 0, \quad \mathbb{E}(u_{t+1} u'_{t+1} \mid \mathcal{C}_t) = \omega_t W_t.$$

Here $\mathcal{C}_t$ contains the design and structural factors; $\omega_t \geq 0$.

Alignment plus local regularity gives $[\theta_0] = [\theta^\circ]$.

Structural-target proposition, Main Section 4; Appendix B. Conditional unit-level sphericity is sufficient. This gives a fixed- N route for the loading class, not consistent recovery of each date's factor.

Interior result: two sources of estimation uncertainty

Let s_t^{prof} be the weighted profiled loading score, G_Q its population sensitivity to the metric, and $\varphi_{Q,t} = \text{vec}(W_t - Q_Z)$ the metric influence contribution.

$$\xi_t = s_t^{\text{prof}} + G_Q \varphi_{Q,t}, \quad \sqrt{T} \hat{\eta}_T = -H^{-1} \frac{1}{\sqrt{T}} \sum_t \xi_t + o_p(1).$$

- The **complete score** ξ_t combines return and metric variation.
- Inverse curvature H^{-1} converts that disturbance into loading movement.
- Same-sample dependence requires their cross-covariances to be retained.

Learning the attribution rule is part of learning the loading class.

Asymptotic linear representation, Main Section 5. Requires regular exact profiling, uniform approximation, score CLT, and numerical score error $o_p(T^{-1/2})$. `vec` stacks matrix columns.

Reports turn the loading class into empirical questions

A **report** is a prespecified smooth statistic of the loading class, unchanged by latent rotations or a different computational chart.

Write $B = \Gamma_\beta$ and let $\mathcal{P}_Q(B) = Q^{1/2}B(B'QB)^{-1}B'Q^{1/2}$ be its whitened-span projector. (The examples below are illustrative, not exhaustive)

Empirical question	Report
How closely does the latent span align with a fixed direction a ?	$\frac{a'QB(B'QB)^{-1}B'Qa}{a'Qa}$
How far is it from a fixed benchmark span D ?	$\ \mathcal{P}_Q(B) - \mathcal{P}_Q(D)\ _F^2$
How much intercept exposure remains?	$\Gamma'_\alpha Q \Gamma_\alpha$

Main Section 6 and Supplement C.2. $a \neq 0$; D has full column rank; $\|\cdot\|_F$ is the Frobenius norm. First-order intervals additionally require nondegenerate report variance.

The same report has the same inference in any admissible chart

Write a scalar report as $\phi(\eta, Q)$, with derivatives ϕ_η and ϕ_Q in the coordinate and vectorized metric directions. The joint delta method gives

$$\sqrt{T}(\hat{\phi}_T - \phi_0) = \frac{1}{\sqrt{T}} \sum_t \zeta_{\phi,t} + o_p(1), \quad \zeta_{\phi,t} = -\phi_\eta H^{-1} \xi_t + \phi_Q \varphi_{Q,t}.$$

- The first term carries loading uncertainty; the second is the report's direct metric response.
- A chart change transforms both terms; their sum is unchanged.
- Apply a heteroskedasticity-and-autocorrelation-consistent (HAC) covariance estimator to the **complete** influence sequence.

Computational convenience need not change the empirical answer.

Invariant-report theorem, Main Section 6; Supplement C. The estimated report is $\hat{\phi}_T = \phi(\hat{\eta}_T, \hat{Q})$ and its target is $\phi_0 = \phi(0, Q_Z)$. Rotation invariance does not imply invariance to the declared metric.

Accounting for metric uncertainty improves interval calibration

Controlled panel experiment: $N = 100$, $L = 12$, $K = 2$, $K_o = 1$; $T = 960$; 2,000 replications. Each panel uses one polished estimate across methods.

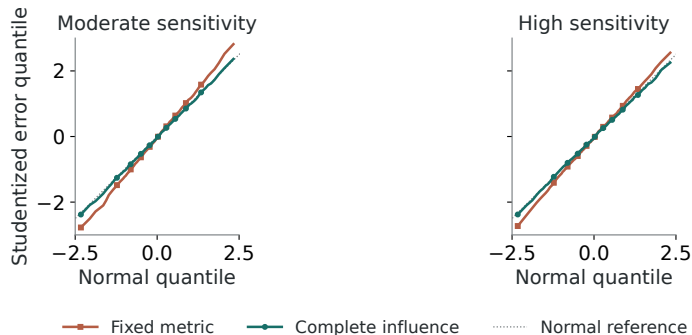
Report	Inference method	SE/SD	95% coverage
Moderate	Metric treated as known	0.838	89.05
Moderate	Complete influence	0.996	94.25
High	Metric treated as known	0.891	91.75
High	Complete influence	1.016	94.85

SE/SD compares the mean estimated standard error with the actual Monte Carlo standard deviation; its benchmark is one. Coverage is in percent.

Coverage gains are 5.20 and 3.10 percentage points at the same estimate.

Main Table 2; Supplement E. Reports: $a' \mathcal{P}_Q(B) a$ for fixed whitened unit directions a , with target 0.5. Paired Monte Carlo SEs of the gains: 0.50 and 0.39 points. Directions selected in a separate calibration; conditional sphericity supports structural centering.

The improvement comes from rescaling uncertainty



Fixed-metric standardization produces overly dispersed errors. The complete influence brings the quantiles closer to the normal reference.

This is an inference gain, not a change in model fit.

Main Figure 2, redrawn from recorded quantiles; $T = 960$, 2,000 panels per report. Studentized error is $(\hat{\phi}_T - 0.5)/\widehat{SE}$. The score-metric cross term is small in this design; the experiment primarily identifies the metfric-variance channel.

An exposure budget creates a second metric channel

The retained intercept may be restricted by

$$b_{\alpha} = \Gamma'_{\alpha} Q \Gamma_{\alpha} \leq \delta_{\alpha}, \quad \delta_{\alpha} \text{ is a declared exposure budget.}$$

At Q_Z , this bounds the population average squared stock-level intercept. It is a substantive restriction, not another normalization.

Let $b_{\alpha,0,T}$ be exposure at the unrestricted population target:

$$\kappa_T = \sqrt{T}(b_{\alpha,0,T} - \delta_{\alpha,T}) \leq 0.$$

- Fixed positive slack: $\kappa_T \rightarrow -\infty$; the constraint becomes inactive.
- Slack of order $T^{-1/2}$: score and metric noise can determine binding.

Near the budget, learning Q moves the boundary as well as the score.

Main Section 7; Supplement D. κ_T measures slack on the root- T scale. The maintained target is feasible; the theorem concerns one smooth boundary with a nonzero normal.

Boundary result: project the joint Gaussian disturbance

Let $(\mathcal{Z}, \mathcal{Z}_Q)$ be the joint Gaussian limits of the complete score and metric error. A local loading movement is $\nu = \sqrt{T}\eta$.

$$\hat{\nu}_T \Rightarrow \nu_\kappa = \underset{\gamma'_\eta \nu + \gamma'_Q \mathcal{Z}_Q + \kappa \leq 0}{\operatorname{argmin}} \left\{ \nu' \mathcal{Z} + \frac{1}{2} \nu' H \nu \right\}.$$

γ_η is the boundary normal; γ_Q is its metric sensitivity.

- The unconstrained movement is $-H^{-1}\mathcal{Z}$.
- Only an infeasible movement is corrected toward the learned boundary.
- A scalar report inherits both projected loading and direct metric movement:

$$L_\phi(\kappa) = \phi_\eta \nu_\kappa + \phi_Q \mathcal{Z}_Q.$$

Learned-boundary theorem, Main Section 7; Supplement D. Subscripts 0 on limiting derivatives are suppressed for readability. Completing the square gives an H -metric projection; its closed form is in backup slide 5.

Unknown slack calls for drift-robust critical values

For each candidate drift $\kappa \in [-\infty, 0]$, simulate the estimated joint Gaussian input, project it, and evaluate the scalar report.

Let $q_\tau(\kappa)$ be the τ quantile of $L_\phi(\kappa)$. For a significance level $\alpha \in (0, 1)$, use

$$q_\alpha^L = \inf_{\kappa} q_{\alpha/2}(\kappa), \quad q_\alpha^U = \sup_{\kappa} q_{1-\alpha/2}(\kappa).$$

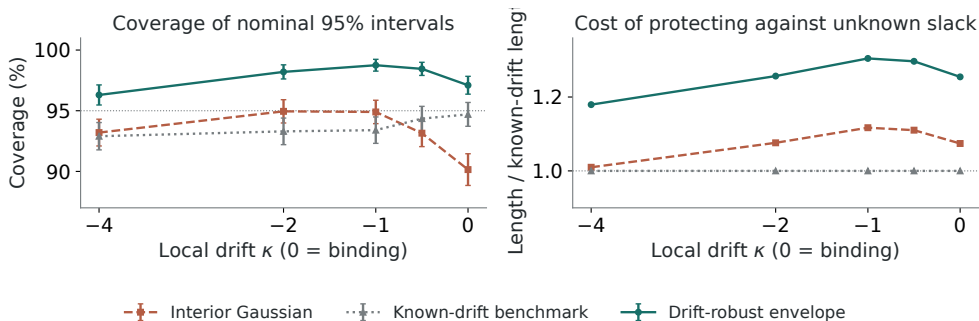
The feasible interval reverses the error quantiles:

$$\left[\hat{\phi}_T - \frac{\hat{q}_\alpha^U}{\sqrt{T}}, \quad \hat{\phi}_T - \frac{\hat{q}_\alpha^L}{\sqrt{T}} \right].$$

One interval protects against unknown local slack.

Uniform coverage theorem, Main Section 7; Supplement D. Under the stated regularity and quantile conditions, limiting worst-case coverage is at least $1 - \alpha$. In this scalar single-half-space law, the envelopes reduce to the interior and binding endpoints.

Protecting against unknown slack has a measurable cost



At binding, coverage is **90.15%** for an interior Gaussian interval and **97.10%** for the envelope; the envelope is about **25% longer** than the known-drift benchmark.

The gain is coverage protection, with an explicit precision trade-off.

Main Table 3 and Figure 3. Local quadratic experiment, not a re-estimated factor panel: $T = 960$, 2,000 replications, 1,499 Gaussian draws. Bars are pointwise 95% Monte Carlo intervals. The known-drift procedure is infeasible in practice.

A flexible framework for empirical exposure questions

1. **Choose the question.** Measure alignment, span distance, block attribution, or another feature of the loading class.
2. **Design your report.** Choose a metric and a prespecified smooth, rotation-invariant statistic. The examples are illustrative, not exhaustive.
3. **Quantify uncertainty.** Retain metric-estimation uncertainty and use inference appropriate to the interior or smooth-boundary regime.

Tailor the report to the empirical question; use a common framework for inference.

Main Section 6 and Supplement C.2. Subject to the stated regularity and nondegeneracy conditions. Boundary inference covers scalar reports under one smooth constraint.

From factor estimation to interpretable inference

- **Starting point:** factor models and IPCA organize common variation and characteristic-dependent loadings.
- **This paper's addition:** an explicit metric-indexed object when latent, observed-factor, and intercept blocks coexist.
- **Inferential payoff:** basis-independent reports with the full uncertainty of the learned metric, including near an exposure boundary.

An explicit exposure comparison, with uncertainty that includes how its measurement rule is learned.

Research presentation based on the September 2026 V2 manuscript and supplemental appendix.

Matched transformation: the algebra behind unchanged fit

Write $B = \Gamma_\beta$, $A = [\Gamma_\delta, \Gamma_\alpha]$, $c = (g', 1)'$, so $h = Bf + Ac$. For full-rank B and $Q \succ 0$, set

$$S = (B'QB)^{1/2}, \quad D = (B'QB)^{-1}B'QA, \quad B_Q = BS^{-1}, \quad A_Q = A - BD.$$

Then $B'_QQB_Q = I_K$ and $B'_QA_Q = 0$. With $f_Q = S(f + Dc)$,

$$B_Qf_Q + A_Qc = B(f + Dc) + (A - BD)c = Bf + Ac.$$

Right multiplication by nonsingular S^{-1} also preserves the raw latent span.

The same fitted index can have a different block allocation.

Main metric-section proposition and Supplement A. This transformation concerns representation equivalence; a budget on Γ_α must be checked separately.

Why covariance alignment removes the fixed-cross-section bias

At structural loadings let $G = (B'W_tB)^{-1}$ and $R = I - W_tBGB'$. Exact profiling gives

$$f^* - f^\circ = GB'u, \quad \tilde{m} = Ru, \quad RW_tB = 0.$$

The loading envelope score multiplies the residual by the fitted factor. Conditional mean zero removes the linear terms; the remaining term is

$$\mathbb{E}[D\tilde{q}_t[\dot{B}, \dot{A}] \mid \mathcal{C}_t] = -\text{tr}\left(R'\dot{B}GB'\mathbb{E}[uu' \mid \mathcal{C}_t]\right).$$

If $\mathbb{E}[uu' \mid \mathcal{C}_t] = \omega_t W_t$, cyclic invariance of trace gives

$$-\omega_t \text{tr}(GB'W_tR'\dot{B}) = 0, \quad B'W_tR' = (RW_tB)' = 0.$$

Main structural-target proposition; Supplement B. Sphericity is sufficient, not necessary. Zero population score plus local positive quotient curvature and slackness selects the structural loading class.

What supports the regular interior approximation?

- **Geometry:** full latent-loading rank, smooth local charts, strictly slack budget, stationary target, $H \succeq c_H I$ for $c_H > 0$.
- **Profiling:** supported datewise latent directions; uniform control of $(B'W_t B)^{-1}$ on the local branch.
- **Sampling:** common moment and dependence bounds, uniform laws for criterion derivatives, and a joint score–metric central limit theorem.
- **Computation:** localization on the selected branch and score residual $o_p(T^{-1/2})$ for first-order equivalence.
- **Feasible covariance:** influence replacement, covariance-tail, and bandwidth conditions for HAC consistency.

Main Sections 3–6; Supplement B–C. Fixed L, K, K_o and strong local identification. These conditions do not follow merely from reporting an optimizer's convergence flag.

Why a chart change leaves the report influence unchanged

Suppose $\bar{\eta} = \tau_Q(\eta)$ describes the same loading class. Let $J = D_\eta \tau$ and $K_Q = D_Q \tau$ in vectorized metric coordinates. Writing $\ell_t = -H^{-1}\xi_t$, the chain rule gives

$$\bar{\ell}_t = J\ell_t + K_Q\varphi_{Q,t}.$$

The same report satisfies $\phi(\eta, Q) = \bar{\phi}(\tau_Q(\eta), Q)$, hence

$$\phi_\eta = \bar{\phi}_\eta J, \quad \phi_Q = \bar{\phi}_\eta K_Q + \bar{\phi}_Q.$$

Substitution cancels the chart terms date by date:

$$\bar{\phi}_\eta \bar{\ell}_t + \bar{\phi}_Q \varphi_{Q,t} = \phi_\eta \ell_t + \phi_Q \varphi_{Q,t}.$$

The net metric coefficient is $\phi_Q - \phi_\eta H^{-1}G_Q$; it vanishes for reports constant along the matched population metric path.

Invariant-report theorem, Main Section 6; Supplement C. Derivatives are evaluated at the population reference. This is invariance to coordinates for a fixed report, not invariance to its economic measurement rule.

Closed-form projection and the two endpoint critical values

Set $\nu^u = -H^{-1}\mathcal{Z}$ and $d_0 = \gamma'_\eta H^{-1}\gamma_\eta > 0$. The half-space projection is

$$\nu_\kappa = \nu^u - H^{-1}\gamma_\eta \frac{[\gamma'_\eta \nu^u + \gamma'_Q \mathcal{Z}_Q + \kappa]_+}{d_0}, \quad [x]_+ = \max(x, 0).$$

Thus a scalar report can be written

$$L_\phi(\kappa) = A - c[V + \kappa]_+, \quad A = \phi_\eta \nu^u + \phi_Q \mathcal{Z}_Q, \quad V = \gamma'_\eta \nu^u + \gamma'_Q \mathcal{Z}_Q, \quad c = \frac{\phi_\eta H^{-1}\gamma_\eta}{d_0}.$$

For each draw, the sign of c fixes monotonicity in κ . The tail envelopes therefore occur at $-\infty$ or 0.

Main Section 7, endpoint-envelope result; Supplement D. One smooth half-space and a scalar linearized report with fixed limiting inputs. Multiple constraints or nonlinear joint rejection regions need separate treatment.

Interpretation and scope of the evidence

- **Fixed N :** the target is a stationary profiled loading class. Covariance alignment supplies a sufficient structural interpretation.
- **Regular reports:** exact alignment, coincident spans, and zero intercept can make first derivatives vanish. First-order Wald intervals are then insufficient.
- **Smooth boundary:** a zero exposure budget has a zero normal; weak curvature and profile rank failure are different nonregular regimes.
- **Numerical evidence:** the Gaussian panel design isolates an interior mechanism; numerical acceptance does not verify an all-date uniform eigenvalue lower bound. The boundary evidence is a local experiment.

Main Sections 6–8 and Supplement D–E. The calibration is not a test of risk premia; portfolio performance requires an additional evaluation design.

Additional interior results and what the comparison identifies

	Moderate report		High report	
	$T = 240$	$T = 960$	$T = 240$	$T = 960$
95% coverage (%)				
Metric treated as known	88.20	89.05	88.15	91.75
Metric variance, no cross term	93.95	94.65	93.40	95.05
Complete influence	93.75	94.25	93.20	94.85

The marginal metric variance matters in this design. The cross term is small here, although the general covariance formula retains it.

Separate a theoretical requirement from the channel a design identifies.

Main Table 2; 2,000 panels per horizon, same point estimates across methods. Both reports have population value 0.5. Monte Carlo standard error near 95% coverage is about 0.5 percentage points.

Selected references

- Bai, J. (2003). Inferential Theory for Factor Models of Large Dimensions. *Econometrica* 71, 135–171. [DOI](#)
- Kelly, B. T., S. Pruitt, and Y. Su (2019). Characteristics Are Covariances: A Unified Model of Risk and Return. *Journal of Financial Economics* 134, 501–524. [DOI](#)
- Kelly, B. T., S. A. Korsaye, S. Pruitt, and Y. Su (2026). Instrumented Principal Component Analysis. Working paper, April 21 version. [SSRN 2983919](#)
- Jensen, T. I., B. T. Kelly, and L. H. Pedersen (2023). Is There a Replication Crisis in Finance? *Journal of Finance* 78, 2465–2518. [Data documentation](#)

Methodological foundations also include two-step extremum estimation, HAC covariance estimation, constrained argmin limits, and least-favorable critical values. Full references and proof-specific attributions are in the paper.